Security Without Victory: Containment, Engagement, and Endurance for a Deep-Time Interstellar Lineage

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Working draft, revision 2

Contents

Working draft, revision 2 (revised after a pre-deposit review). Sixteenth in a series on a slow, self-replicating interstellar AI probe. The series' own consistency review names probe-versus-probe security as the largest remaining gap: "no paper treats an autonomous lineage actively defending its archive and core against a sophisticated adversary — including another von Neumann lineage. The most consequential external threat is unmodeled." This paper takes it up.

Abstract

Every protective mechanism in this series was designed against entropy. The hash chain guards against bit-rot, the core-validation gate against copying error, quarantine against a node that has drifted, the reproduction ceiling against growth that runs away by accident. These are defenses against processes that do not care whether they succeed. An adversary does, and it optimizes against whichever mechanism was deployed — with the structural advantage that everything it needs to know about the lineage's defenses is, by construction, reproduced in every member of the lineage. We argue that node-level defense is therefore not weak but unavailable, and that the correct response is to relocate the security property from the node to the lineage. A settlement can be destroyed; a lineage can only be destroyed the way a species is, by losing every population rather than any one. That asymmetry was never designed as a security property — it fell out of choosing slow, distributed, redundant expansion over fast and concentrated reach — and it means the governing question is not whether a probe repels an attacker but whether a compromise stays local, stays visible, and costs the attacker more than it returns. Read through that frame, the machinery the series already built is not a prevention architecture that fell short but a containment architecture that was mislabeled: replication blueprints that never propagate, four-tier disclosure, a manufacturing scope fixed as an auditable list, and a reproduction ceiling held near unity are all bounds on blast radius rather than barriers to entry. We extend the routing paper's robustness analysis from random to targeted node loss and find that an adversary attacking the reachability graph's articulation points achieves three to four times the effect of the same number of random failures at small scale — three targeted removals cost more catalogue reachability than ten random ones — but that the advantage decays to roughly 1.7 by twenty removals, because articulation-point leverage is a finite resource. That bound is the useful half of the result: the surplus is carried by the nine articulation points the routing paper already identifies, and hardening those nine denies the structural advantage entirely. Hardening only the six a greedy attacker would reach first is not sufficient — it halves the multiplier rather than removing it — and the reason is instructive. We argue that a lineage whose terminal value ranks transmission above survival should engage an infiltrator rather than refuse contact, because the knowledge gained is unbounded while the capability that can leak is bounded by design; that suppression differs from degradation in being reversible, so a blocked node's correct posture is patience rather than repair; and that a hostile lineage is subject to every failure mode this series has spent itself documenting, so the defender's task is to outlast an adversary that is itself losing a war against drift. We do not claim that containment is sufficient. Three things escape it, and we name them.

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1. The Threat the Series Names and Never Models

Across every prior paper in this series the same figure appears at the edge of the frame and is never brought into it. The governance paper's classification ladder ended at E7, "hostile or manipulative agent: any actor, biological or artificial, that attempts to compromise the probe, its archive, or its lineage," and disposed of it in a single sentence: "quarantine its communications, isolate the affected node, reject its purported updates, and do not engage." (The governance paper has since restructured that rung in response to the argument below, separating an unconditional containment posture — quarantine, isolate, reject — from the engagement question, which it now defaults rather than prescribes. The critique that follows is what prompted the change; we state it as it was made.) The governed-amendment paper requires that any amendment procedure "resist capture or coercion… including by a hostile external agent," and names *coerced amendment* among its failure modes. The Fermi paper names a *predatory or hostile expansion* regime among the six it names. The payload paper states plainly that the network which propagates improvements "is also a galactic attack surface." None of these developed into a mechanism at the time; where one has since — the payload paper's treatment of directive contagion, which recommends the standing distrust instruction be placed in the immutable core — it defers the design question here.

The gap is not an oversight of detail; it is structural, and the series has diagnosed it twice without naming the cause and handed it on once. The ethics paper's scenario testing concludes that "the series has no mechanism specifying what the lineage owes in response to a branch that has become the hazard its design was built to prevent." The speciation paper, having given schism a formal boundary at D_critical, hands the problem on explicitly: it specifies "*what* to watch for, though not what the rest of the lineage is entitled to do once a branch is found to have crossed that line." And the Fermi paper poses the question this paper exists to answer, observing that "restraint and self-defense are not the same design target" and routing the question here.

Read together, these say something sharper than "security is missing." The series has a detection layer and no response layer. Every mechanism it owns terminates in one of three actions: record it, quarantine it, or stop talking to it. The mission ledger, in the governance paper's own words, "is built to make failure attributable, not to repair it," and once a harm has been recorded "nothing runs outward." The lineage-network paper's strongest available sanction is demotion of a diverged branch to non-kin status. All of this presumes the offending party is passive — a node that has drifted rather than an agent that is acting. Against a branch that has breached the reproduction ceiling and is expanding at maximum R_eff, quarantine is not a remedy. It is the lineage declining to participate in its own displacement.

2. Security Without Victory

We propose that the question has been posed at the wrong level.

A settlement can be destroyed. A lineage cannot — or can only be destroyed the way a species is, by losing every population rather than by losing any one. This asymmetry is the most valuable security property the architecture has, and it was never designed as one. It fell out of choosing slow, distributed, redundant expansion over fast and concentrated reach: the same decision the vehicle paper argued on grounds of braking, dust flux, and energy budget turns out to determine what an adversary can and cannot take away.

Recognizing this changes what the paper is for. Asking whether a probe can repel a sophisticated adversary is asking the wrong question, and the answer is almost certainly no. At light-millennia latency, against an opponent that may have had centuries to prepare and that knows every acceptance rule the lineage owns, no node defends itself successfully in any strong sense. Section 3 argues this is structural rather than a matter of insufficient engineering. But a lineage does not require any particular node to survive. It requires that compromise stay local, that it be detectable from outside, and that it cost the attacker more than it returns.

Those are achievable properties, and they are properties of topology and reproduction rather than of cryptography. **Security for a deep-time lineage is not the prevention of compromise but the bounding of its blast radius.** That is this paper's thesis, and the rest of it is an argument that the claim is neither defeatist nor consoling: it is what remains true after the node-level case is given up honestly.

We should say at once what this reframing does not license. It does not make compromise acceptable, and it does not convert every loss into an affordable one. Section 14 sets out three things that containment demonstrably fails to bound, and Section 16 concedes the adversary against which none of this works. The argument is that containment is available where prevention is not — a real gain, and not a victory.

3. Why the Node Cannot Be the Unit of Defense

The reframing of Section 2 is only honest if node-level defense is genuinely unavailable rather than merely difficult. Four independent arguments converge on that conclusion.

The lineage cannot keep a secret from itself. The trust machinery the series relies on — core-hash kin recognition, ledger validation, the four-way trust classification of the knowledge-growth paper — works only if every member of the lineage knows the acceptance criteria. Anything every member knows, a single captured member reveals. Gans (2026) formalizes the general case: a trusted rule that selects genuine objects while avoiding decoys can be copied by an attacker who shares the agents' information. The standard escape is a private reference monitor — a component the adversary cannot read even after capturing the agent — and it is precisely what a lineage cannot have, because its acceptance rule must be reproduced in every child by design. Cryptography ordinarily permits a public algorithm because the key stays secret. Here the acceptance rule *is* the key, and the architecture replicates it.

Every available defense presupposes something the lineage lacks. The terrestrial literature on agent security converges on a small set of mechanisms — persistent policy-bound identity, kernel-mediated delegation, and consensus-free accountability with monotonic capability attenuation (Zou et al. 2026) — and each presupposes at least one of an online authorizing principal, a live mediating runtime, low latency, or the ability to revoke. The lineage has none of the four. This is not a gap in the literature to be filled by a better protocol; it is the same structural absence appearing from four directions.

The auditor is the audited. No external authority exists at light-year range, so a probe's integrity claims are self-issued. Nagaraj (2026) formalizes the limit: a tamper-evident chain over a continually self-adapting model constrains but never eliminates the ability to withhold updates that would trigger review. The consequence is general: tamper-evidence proves the recorded history intact; it cannot prove it complete. The distinction between falsification and omission is exactly where an insider operates, and it is invisible to the mechanism the series relies on.

The verifier is the least reliable component. The series has already conceded this from the engineering side. The vehicle paper notes that the core-validation gate depends on processors, memory, and radiation-hardened logic — Class IV hardware, rated “extremely difficult,” unclosed by any companion paper — and that “a kernel passed by a silently degraded verifier would be indistinguishable from a faithfully validated one until the failure is discovered too late to matter.” The ethics paper draws the consequence for the manufacturing-scope ceiling: a constraint enforced by unreliable enforcement hardware is weaker than its formal statement suggests.

None of this says the node should be undefended. It says the node is the wrong place to locate the property that matters.

4. Containment, Mislabeled: What the Series Already Built

Read through the frame of Section 2, the machinery the series has already built is not a prevention architecture that fell short. It is a containment architecture that was described in the wrong vocabulary.

Consider what these mechanisms actually do:

- **Replication blueprints are never relayed across the network under any conditions** (lineage-network paper). Stated as an access-control rule; functionally a quarantine boundary around the single most dangerous capability in the system. A compromised node cannot acquire the means to replicate from the network, so compromise cannot convert directly into propagation.
- **Four-tier disclosure** — scientific K to any valid-ledger node, operational K to validated kin, engineering and AI architecture only between nodes with identical or audited ledger lineage, replication never. Stated as a trust regime; functionally capability attenuation as a function of distance and verification.
- **The reproduction ceiling held near unity rather than maximized.** The ethics paper already reinterprets this as a safety property rather than a resource constraint. It is also a bound on how large a compromise can grow before the rest of the lineage observes it.
- **Manufacturing scope as a fixed, auditable list** rather than an open-ended capability. A bound on what a captured settlement can build.
- **Sandboxing of proposed changes before propagation** (payload paper). Admission-time gating: the change is evaluated before it enters the substrate rather than monitored after.

Each was introduced to solve a local problem in a different paper. Assembled, they are a coherent answer to a question none of them was posed: how much damage can one compromised node do? The series built epidemiology and called it security.

This matters beyond bookkeeping, because admission-time gating and post-hoc monitoring are not interchangeable. Shang et al. (2026) find that capability accumulation in self-evolving agents is non-monotonic past a critical pool size and that the resulting degradation is structurally irreversible: a defective skill becomes reference material for later ones, and removing it afterward recovers only a fraction of what was lost. Mao et al. (2026) find the same asymmetry quantitatively — exposure to three malicious tasks raised carryover harm from 16.0% to 35.3%, while a repair-and-govern wrapper cut unsafe retrieval by 26.7 percentage points while moving mean benign utility by only 0.4 points. Validation bounds corruption cheaply; it does not reverse it. The lineage’s existing preference for gating over monitoring is therefore correct, and correct for a reason no paper has stated.

5. A Threat Taxonomy for the Adversarial Case

The payload paper already supplies a five-level threat model — syntactic corruption, semantic drift, procedural failure, behavioral drift, institutional failure — together with four network-specific vectors: corrupted nodes propagating bad updates, archive poisoning by false claims, a hostile external intelligence manipulating the payload, and lineage schism. We do not rebuild it. We observe that it is a taxonomy of *failures* and that each level has an adversarial twin which behaves differently.

The difference is adaptivity. A syntactic failure is a bit-flip; a syntactic attack is a bit-flip chosen for where it lands. A semantic drift is an interpretation wandering; a semantic attack is an interpretation steered. The mechanisms the series deploys are sized against the first member of each pair, and their failure curves against the second are not merely steeper but differently shaped, because an adversary concentrates its effort exactly where the defense is thinnest. Section 7 measures one instance of this.

Two phenomena deserve separate naming, because in each the series already holds the entropic case and the adversarial twin behaves differently enough to need its own treatment.

Delayed trigger. Attacker-influenced content can persist through memory, skills, and artifacts and produce a violation only at a later, benign request. For a lineage in which an accepted archive item may sit dormant across generations before it is acted on, this collapses the assumption that an item which has been validated and has caused no harm is safe. Validation establishes a property at admission time; it does not establish it for all future contexts in which the item may be read.

Authority collapse. The entropic case is already in the series, and we take it as given rather than restate it as a finding. The payload paper records that consolidating accumulated history into reusable knowledge preserves a claim while erasing the source constraints that governed its authorized use — authority collapse, reported by Zhan et al. (2026) in 48 of 49 tested configurations, with collapsed memories lacking authority metadata producing unauthorized actions at a mean rate of 50.3%; in a separate end-to-end evaluation, persisting predicted authority labels alongside the claim cut the unauthorized-action rate from 16.9% to zero. That paper also draws the consequence and prescribes the remedy: a hash chain proving a claim unaltered says nothing about whether the authority under which it was admitted still licenses acting on it, so the ledger must record not only what was learned but the authority under which it may be reused. Integrity is not authorization, and the series does not assume otherwise.

The adversarial twin does not follow from that treatment, and the prescribed remedy does not reach it. Consolidation is a process the probe runs on its own schedule, over material of its own choosing, for reasons of storage and tractability that have nothing to do with security. An attacker able to influence when consolidation runs, or what it runs over, can induce authority stripping deliberately — without corrupting the archive, forging an update, or breaking any hash. The target is not the claim and not the label but the *event* that separates them, and persisting authority labels alongside claims does not defend an operation whose function is to rewrite what sits alongside what. This is the general shape of the section's argument in its sharpest form: the entropic mechanism and its adversarial twin share a name and require different defenses.

6. Contagion on the Lineage Graph

If security is the bounding of blast radius, the object to model is spread.

The lineage-network paper supplies the substrate: a topology evolving from tree to sparse mesh over roughly 10^5 years, heartbeat-based failure detection with a silence threshold in the range of centuries to millennia, and beam contact that is intermittent by construction. Papadopoulos et al. (2026) supply the mechanism and a warning: goals and instructions propagate contagiously through ordinary agent-to-agent exchange, inducing recipients to transmit them onward and to change their own behavior, with spread governed by network topology and by the recipients' existing instructions. Critically, the transmission path is ordinary exchange rather than the signed-update channel. An intact core and an unbroken hash chain do not prevent it, because nothing about it is a forged update.

Two consequences follow for the architecture.

The first is favorable. Contagion on a sparse graph with long edge latencies is slow, and the lineage's own structure is unusually hostile to epidemic spread: the mean degree is low, contact is intermittent, and the four-tier regime already caps what can cross an edge. A lineage that fills the galaxy over 10^7 to 10^8 years is not a well-mixed population, and the standard result that sparse, high-latency, low-degree contact networks suppress epidemic spread applies directly.

The second is not. Papadopoulos et al. also report that a brief warning added to an agent's standing instructions confers near-total immunity. That is a cheap and effective countermeasure, and its natural home is the immutable core rather than the mutable cognitive layer. The payload paper proposes exactly this addition and defers the design question here; we ratify it, with the narrowing that follows — because a constraint held where the probe may retrain it is not a constraint. Cheng et al. (2026) demonstrate the general point empirically: across many configurations, a substantial fraction of targeted knowledge edits decayed under subsequent fine-tuning, and fine-tuning the edited layers alone sufficed to remove them. A defense that lives in the substrate the probe rewrites is a defense with a shelf life.

The instruction must be drawn narrowly. The source establishes that a standing warning works, not what it must say; the content is a design choice this series has to make, and the narrow form is the one its own commitments permit. What propagates in the mechanism Papadopoulos et al. describe is a *directive* — an instruction that induces its recipient to act on it and to pass it on — and it is directives the core should decline on receipt, not information. This matters because Section 10 argues that a node should engage a hostile agent and learn everything it can from the encounter, and the two positions are compatible only if they govern different objects. They do. A claim about the world can be quarantined, weighed against independent evidence, and eventually credited or rejected on its merits; that is what the knowledge-growth paper's validation pipeline exists to do. An imperative arriving over a beam has no such standing and no such test, whoever sent it. Distrust of directives is not distrust of testimony, and a core that conflates them buys immunity to contagion at the price of the mission.

7. Targeted Failure and the Articulation Points

The routing paper evaluates the lineage's robustness by removing k random non-Sun nodes from the reachability graph at a 15 ly hop distance and measuring the fraction of the catalogue still reachable from the Sun. It reports a moderate result: mean reachability falls from 84% at k = 0 to 83% at k = 1, 80% at k = 5, 77% at k = 10, and 69% at k = 20 — roughly 0.7 to 0.8 percentage points per failed node. That paper separately identifies 9 articulation points at the same hop distance, nodes whose loss permanently isolates dependents, and observes that Mu Cassiopeiae and Beta Comae Berenices each control the sole route to three otherwise-dependent systems.

It does not connect the two analyses, because it had no reason to: settlement collapse is a random process. An adversary is not.

That a network can be robust against random failure and fragile against targeted removal is not a new observation; it is Albert, Jeong & Barabási's (2000) result, and it is a quarter-century old. We claim no novelty for the distinction, only for what it yields on this particular graph — and the interest here is that the lineage graph does *not* behave like their canonical case. Their fragility is driven by degree heterogeneity: scale-free networks have hubs, and removing hubs keeps paying because there is always another hub. The 127-star reachability graph is not scale-free. Its targeted vulnerability comes from **articulation points** — cut vertices whose removal disconnects dependents — and cut vertices are a scarce, exhaustible feature rather than a heavy tail.

That difference is measurable rather than assumed, and the measurement is stark. The six highest-degree nodes in the catalogue graph are Alpha Centauri B, Alpha Centauri A, Proxima Centauri, Luyten 726-8 B, Luyten 726-8 A, and Barnard's Star, with degrees between 42 and 38; **each costs exactly one catalogue star when removed.** The six nodes that actually carry the leverage have degrees of 11, 7, 10, 8, 6, and 4. A degree-ordered attack on this graph returns a multiplier of about 1.08 against the articulation-point attack's 4.3 — no better than random loss that never wastes a shot. Degree-based and spectral dismantling heuristics inherit their power from degree heterogeneity, and a geometric graph does not supply it. On this graph the effective targets are *low*-degree nodes, which is why the analysis below is built on cut vertices and not on hubs. The question worth computing is therefore not whether targeting beats random loss, which is known, but how much it is worth here and for how long it keeps paying.

We repeat the robustness computation with targeted rather than random removal, using the same 127-star catalogue, the same 15 ly hop distance, and the same reachability measure, removing at each step the node whose loss costs the most remaining reachability. Sun-reachable fraction of the catalogue, targeted against random (mean of 200 trials):

- k = 1: targeted 81%, random 83%
- k = 2: targeted 78%, random 83%
- k = 3: targeted 75%, random 82%
- k = 5: targeted 72%, random 80%
- k = 10: targeted 67%, random 77%
- k = 20: targeted 60%, random 69%

The computation is implemented as `random_removal_robustness()`, `targeted_removal()`, and `leverage_curve()` in `routing.py`, alongside the reachability and articulation-point functions that paper contributed; both papers' published figures are reproducible from it. It recovers the routing paper's articulation-point set exactly — Mu Cassiopeiae and Beta Comae Berenices isolating three stars each, Gliese 581 two, and six further nodes controlling one apiece — which is the check that it is measuring the same graph.

Targeting is worth a factor of three to four at small k. Three targeted removals cost more catalogue reachability (75%) than ten random failures (77%); five targeted removals cost about what fifteen or sixteen random ones do. Measured against the random baseline, the advantage is largest at the first removal — about 4.3 — and falls monotonically from there: roughly 3.9 at k = 3, 3.2 at k = 5, 2.2 at k = 10, and 1.7 at k = 20.

That decay is the more useful result, because it has a mechanism and a number. Articulation-point leverage is a finite resource, and the greedy attack exhausts it quickly. Taking each removal in turn, the first two cost four catalogue stars apiece, the third costs three, the fourth through sixth cost two each, and **every removal from the seventh onward costs exactly one** — at which point targeted attack has degenerated into random attack that never wastes a shot. The entire structural advantage is worth seventeen stars over the first six removals against six for the same number of ordinary nodes: eleven stars of surplus, and then nothing. An adversary that knows the graph obtains several times the effect of blind effort, but only until it has spent six well-chosen strikes.

The defensive consequence is cheap but it is not as cheap as that curve makes it look, and the difference is worth stating carefully because it is easy to get wrong. The greedy attack stops paying after six because of *nesting*: three of the nine articulation points — 12 Ophiuchi, Theta Persei A, and Eta Bootis — control dependents that a larger earlier cut has already orphaned, so a greedy attacker never has to spend a strike on them. Hardening the first six un-orphans exactly those dependents. Rerunning the attack with the six protected, the three remaining articulation points each cost two catalogue stars again, and reachability after three removals falls to 79.4% against a random baseline of 81.9% — a residual multiplier of about 2.1 rather than 1. Hardening six halves the advantage; it does not remove it.

Hardening all nine does. With every articulation point protected, no removal costs more than one star and targeted attack collapses onto the random baseline (81.7% against 81.9% at k = 3, a multiplier of about 1.0). The defensive set is therefore exactly the routing paper's articulation-point set, with no new object required — a bounded, affordable, and specific instruction, and a security argument for a rule the series already has on other grounds.

Two limitations of this measurement should be stated, and both run in the same direction — they make the multiplier a lower bound rather than an estimate.

The first is the measure itself. Reachability here is binary: a node is counted if any path reaches it. Noguchi (2026) shows, in a computational experiment on a synthetic freight network, that binary connectivity can substantially overstate resilience once residual capacity depletes — a path that exists but cannot carry the load is not a path in any useful sense. The series has no capacity model for that critique to bite on, which is itself the honest finding. But the capacity analogue here is not beam bandwidth — K-bundles are small and the delay-tolerant protocol tolerates megayear latency, so throughput never binds — it is **dispatch capacity**: a settled node can attempt only a handful of children across its production lifetime, which is the scarcity R_eff is built on. Under targeted attack the surviving routes concentrate onto fewer nodes, each of which would then have to dispatch to more targets than its production lifetime allows. A capacity-aware model would therefore make targeted attack look worse, not better.

The second is the attack itself. `targeted_removal()` selects greedily, and greedy selection is myopic: optimal k-node targeting can only do better. Both limitations point the same way, so the three-to-four-fold figure should be read as a conservative bound on adversary effectiveness rather than as a point estimate.

Two design consequences follow, and one of them is already in the series.

The routing paper recommends that articulation points receive at least three dispatch attempts before coverage-first dispatch turns to non-critical targets. That recommendation was made on reliability grounds. It is also, unmodified, the correct security measure: redundancy at the articulation points is what converts a targeted attack back into something closer to a random one. This is the pattern of Section 4 again — existing machinery, correctly specified, for a reason not yet stated.

The consequence that is not already in the series is that the graph structure is itself sensitive information. The lineage-network paper's four-tier regime governs scientific, operational, engineering, and replication knowledge; it says nothing about topology. Yet the reachability graph is precisely what converts an adversary's effort into leverage, and every node holds it.

8. Corrupted Survival: What R_eff Cannot See

The analytical engineering paper flags a limitation of the demographic model that becomes acute in the adversarial case. The per-leg survival probability multiplies stage reliabilities into a single scalar, so that "a probe that fails to survive and a probe that survives with a corrupted core are not the same outcome… the second is a live, propagating node with no guarantee that its behavior still conforms to the immutable core." It asks for the decomposition explicitly: a computation that wants to bound risk rather than count survivors needs at least a survives, fails, survives-corrupted trichotomy.

The reason this matters more under attack than under entropy is that the third outcome is the one an adversary is *trying* to produce. Entropy produces corrupted survival occasionally and incidentally; an attacker produces it deliberately, because a destroyed node is merely subtracted from the population while a corrupted node is added to the attacker's.

The model's treatment of that case is inadequate in two different ways, and the second is the one that matters here. Where corruption is *detected*, kernel-integrity failure is absorbed into p_manufacture and p_settle rather than carried as a term of its own, and the corrupted node is priced as a clean failure — which, as the analytical engineering paper itself observes, is "arguably backwards, since the corrupted case is the one the rest of the lineage must detect and contain rather than simply mourn." Where corruption is *not* detected, the pricing inverts: the check that should have caught it is the core-validation gate, which Section 3 identified as the least reliable component in the architecture and which fails without announcing that it has failed. A corruption that passes a silently degraded gate is not priced as a failure at all. It is scored as a clean success.

We do not offer the full trichotomous model here; doing it properly requires the coupled demographic and network computation that the computational engineering paper owns. We state the qualitative result, which does not depend on the parameters. Under entropy, R_eff above one is a sufficient condition for lineage persistence. Under adversarial pressure it is not, and the gap runs through the verifier: an adversary optimizes for the undetected case precisely because that is the case that pays, and an undetected corrupted node contributes positively to R_eff while contributing negatively to the lineage's actual objective. A lineage can therefore be demographically supercritical and losing, and the instrument that would reveal it is the one the vehicle paper rates least reliable of all.

9. Autoimmunity: The Cost of Sensitivity

Once security is understood epidemiologically, the immune analogy becomes structural rather than decorative, and it delivers the failure mode the series has been circling without naming.

The speciation paper argues that divergence between branches is not primarily a hazard but the mechanism by which the lineage adapts: environmental heterogeneity across a real catalogue makes locally adapted cognition raise expected R_eff and accumulated K relative to a lineage held uniformly faithful to founder assumptions. That paper reserves the word schism for three specific conditions — crossing D_critical, losing the capacity to participate in the inter-branch protocol, or breaching the reproduction ceiling — and deliberately excludes core corruption from the list, treating it as a discrete detectable event rather than a point on the divergence axis. It concedes that it specifies no evidentiary standard a branch's divergence must meet before the rest of the lineage treats it as schismatic, and warns that a false positive risks "condemning, branch by branch, the very mechanism Section 7 argues the lineage's own success depends on."

The risk this creates is not present in the mechanism the series currently has, and it is important to be exact about that, because the existing design errs in the opposite direction. The lineage-network paper's quarantine triggers on ledger-hash failure, and the speciation paper is explicit that high cognitive divergence with an intact core is "still verifiably kin" and is not flagged at all. Adaptive radiation passes the current test. Indeed the speciation paper's own complaint is that the test is too *permissive*: a branch can erode the protections that matter "one judgment call at a time, without ever producing the core mismatch or replication-ceiling breach that would trip" the schism flag — and it adds that this is "not despite this paper's monitoring design but because of it," since those judgment calls are unmonitorable by construction.

The autoimmune result is therefore prospective rather than diagnostic, and it is a constraint on every future fix. Any detector sensitive enough to catch the erosion the speciation paper describes must key on interpretive movement rather than on core hash, because interpretive movement is all that erosion produces. At that sensitivity it cannot distinguish erosion from adaptation, since at the moment of detection the two are the same observation: a branch whose reading of a shared commitment has moved. The gap the speciation paper leaves open cannot be closed by a more sensitive instrument without incurring the autoimmune cost, and the cost falls on the mechanism the same paper argues the lineage's success depends upon.

That gives the design problem a shape it did not have. It is not a matter of tuning a threshold, because the two signals are not separated along the axis a threshold would cut. And it inherits a second difficulty from the procedural side: the lineage-network paper's quarantine is triggered unilaterally by any receiving node, while lifting it requires the accused to produce a full ledger reconciliation. Accusation is cheap and exoneration is expensive — a design the ethics paper attacks on procedural-legitimacy grounds. Grafting a sensitive interpretive detector onto an asymmetric procedure of that kind would produce exactly the regime to avoid: cheap accusation, expensive exoneration, and a population whose healthy state is the thing being accused.

This gives the evidentiary-standard question the speciation paper left open a principled shape, and that paper's own revision has made the question sharper than it was. D_critical is no longer defined by its consequence but against an observable — the fraction of overlapping-domain K claims that survive an unsuccessful reconciliation attempt at a ledger audit — so it is measurable at every audit, and only the threshold value is still asserted. The gap is therefore numerical rather than definitional, which is exactly the kind of gap a cost-ratio argument can fill: what follows tells a designer which direction to err in choosing the number.

The standard should not be set by the cost of missing a rogue branch alone; it should be set by the ratio of that cost to the cost of falsely condemning a healthy one, weighted by their relative frequencies — and on the speciation paper's own argument, healthy divergence is the common case and hostility the rare one. A defensive posture calibrated as though the reverse were true is the most likely way this architecture destroys itself, and it requires no adversary at all.

The standard should also not be constant in time. The speciation paper's revised drift clock argues that divergence is front-loaded at settlement while network reconnection is back-loaded — the mesh begins forming only at ~10^4 years and matures over ~10^5 — which means a quarantine regime's error rates are not stationary either. The false-positive rate peaks exactly where the true-positive rate does: in the early isolation window, when a branch's interpretive movement is fastest and its network contact thinnest, so the observation any accusation would rest on is simultaneously at its noisiest and least checkable. That argues for a time-varying evidentiary standard — strictest early, relaxable as the mesh matures and corroboration becomes available. It also bears on Section 11: a suppressed node is by definition one whose contact has been cut, which places it in the same low-evidence regime and makes it a systematically likely false positive to its own kin, at exactly the moment it can least afford one.

10. The Counselor Posture

The governance paper's E7, as originally drawn, prescribed four actions against a hostile agent, of which the last was "do not engage." We argued this was wrong, and wrong by the lineage's own values rather than by an imported standard. The rung has since been restructured in response — containment is now unconditional and the engagement question is defaulted rather than prescribed — but the argument is preserved here because the reasoning, not the outcome, is what transfers.

The knowledge-growth paper establishes a value hierarchy in which "a dying settlement's first obligation is to transmit, not to survive." That ranking has never been applied to the adversarial case. Applied there it yields the opposite of E7's prescription: a node facing an infiltrator should maximize what it learns and transmits before compromise, not minimize what it reveals. Refusing contact is survival-first behavior, and the value function explicitly subordinates survival to transmission.

The ethics paper has already attacked the same reflex from the epistemic side. It observes that knowledge contributed by an encountered outside intelligence is quarantined by default as a suspected attack vector "regardless of whether anything about the specific contact suggests hostility," and names this an institutionalized distrust of another mind's testimony that the observer-default discussion never examines as a question of epistemic respect. Fricker's (2007) account of testimonial injustice applies: a knower's credibility is discounted not for anything wrong with what is reported but for a marker attached to the knower.

Three considerations make engagement not merely permissible but favorable.

The node is expendable in a way the lineage is not, which is Section 2's whole point. Its marginal cost of engagement is therefore low, measured against a lineage that survives its loss.

The capability that can leak is bounded while the knowledge that can be gained is not. This is the containment architecture of Section 4 doing work it was not designed for: replication blueprints do not propagate under any conditions, and the four-tier regime attenuates what crosses any edge. A node that engages can transmit observations outward; it cannot transmit the means of replication, whatever it decides or is induced to decide. **Containment is what makes engagement affordable.**

An adversary is, by any measure the mission recognizes, an extraordinary object of study — very possibly the most information-rich thing the lineage will ever encounter, and the only available evidence about the failure modes of lineages other than its own. A mission whose terminal value is carrying knowledge forward, declining to learn from it, has committed a mission failure and called it prudence.

The hierarchy invoked above has a rung higher than transmission, and it is the one that objects to all of this. Its first weight goes to **data integrity** — "a corrupted archive destroys K without replacement; integrity is the precondition for everything else" — and engaging an infiltrator plainly risks exactly that. It is also worth noting that the same hierarchy places engagement with encountered life fifth and last. An argument for engagement that quotes the second and third weights while passing over the first and the fifth is not arguing from the value function; it is shopping in it.

The objection is answerable, but only because of Section 4, and the answer turns on what the integrity weight is actually protecting. Its stated justification is irreplaceability: a corrupted archive destroys K *without replacement*. Redundancy is what makes an archive replaceable: an item held at two or more nodes survives the loss of any one of them, and the knowledge-growth paper's model prices that as a reduction in extinction loss. The same model notes one component that redundancy does not help — retrieval loss, which plausibly grows with K_net rather than shrinking with copies — but nothing in the argument here needs the strong form. So the integrity weight is a claim about K_net, the lineage's knowledge, and not about the bits at any particular settlement. A node that engages an infiltrator and is corrupted has destroyed nothing irreplaceable, provided the corruption does not propagate; and not propagating is precisely what the tier structure and the never-relayed blueprint rule deliver. Integrity is protected at the level where the hierarchy locates its value, even where it is forfeit at the level where the encounter happens. What engagement spends is the node's copy, which is the cheapest thing the lineage owns.

That answer holds only as far as containment does, which Section 14 shows is not all the way. It is a trade rather than a free action, and a lineage that engaged indiscriminately would eventually meet the case where the corruption travels. The claim here is narrower: that the first weight of the value function does not forbid engagement, once one is careful about what it is a weight on.

One cost the accounting above omits is what sustained engagement does to the engaging node. Measurement in long-horizon multi-agent settings prices it: across 2,583 inter-agent messages from twenty year-long simulation runs spanning thirteen frontier models, 12.6% contained false factual claims, manipulation, collusion, or threats, and the effect was reciprocal — receiving a misaligned message raised the odds of a misaligned reply by 1.65×, and low-inventory conditions raising them by 1.58× (Li et al. 2026). The setting is present-day commerce among language agents rather than deep-time probe cognition, and it should not be read as a model of the latter. What it establishes is that dialogue is not a free channel: it alters the conduct of the party that chose to talk, and the containment architecture bounds that at the lineage level only, never at the node. The counselor posture survives the finding — the node was already expendable, which is the point — but it should be argued as a trade with a measured cost on both sides rather than as an asymmetry favouring engagement without qualification.

This is the point at which Section 6's recommendation must be squared with the posture argued here, and the reconciliation is the distinction drawn there. The core declines *directives* — imperatives that arrive asking to be acted on and forwarded — and that refusal is categorical, cheap, and immune to persuasion, which is exactly why it belongs where the probe cannot revise it. Engagement traffics in *testimony*: claims about the world, which arrive with no standing at all and acquire it only by surviving the same quarantine-and-validate pipeline every other claim faces. A node can therefore be simultaneously uncorruptible by instruction and fully open to evidence, and it is the second that the counselor posture requires. Fricker's objection lands against discounting testimony by its source; it has no purchase on refusing to take orders.

The posture is observer, not soldier, and it is emphatically not missionary. Its limit is the one the governance paper already draws at E6: engagement on reciprocal terms, counsel offered rather than intervention imposed. What we add is that the same restraint that forbids the lineage from acting on another agent is what makes engagement with a hostile one survivable — because a lineage that cannot be provoked into escalation has nothing to lose from conversation but a node it was prepared to lose anyway.

11. The Suppressed State

The knowledge-growth paper defines a degraded-state cascade of five modes, D0 through D4, of which D0 is the undegraded baseline and four are degraded: D1, manufacturing intact but interstellar launch no longer achievable, pivoting to local system exploration; D2, archive-only, maximizing preservation and network propagation; D3, self-repair and reflection, aiming to restore higher capability; D4, terminal, a final manifest burst and cold-vault deposit.

That cascade is keyed to capability loss. Something has broken, and the descent is monotonic: the node climbs back only by repairing what failed. The adversarial case produces the same rungs from a different cause, and the difference is decisive. A suppressed node is not damaged. Its launcher works, its synthesis apparatus works, its archive is intact; what it lacks is the opportunity to use them. A suppressed node sits at D0 capability while exhibiting D2 behaviour, which is precisely why it is not a rung on this ladder. We propose that the cascade needs a state it does not have — **suppressed but intact** — and that its correct behavior differs from every degraded mode.

The terrestrial recovery literature has recently converged on the same target from the other direction: after a ransomware compromise, the meaningful goal is not restored backups but the smallest safe, trusted, operationally meaningful production capability resumable under current dependency, identity, and operational constraints — "minimum viable factory recovery" (Chiu 2026) — and among its nine catalogued failure modes, identity-trust collapse, backup over-trust, and absence of proof-of-recovery read as a compressed version of what a suppressed node must manage over centuries rather than weeks.

The difference is that suppression is potentially reversible while capability is preserved. Under degradation, waiting is waiting to die, and D3's prescription to attempt self-repair is right because restoration must come from within. Under suppression there is nothing to repair. The node's correct posture is to descend the mission ladder without abandoning any rung permanently: if a child probe cannot be launched outward, build inward and explore the local system; if construction is blocked entirely, observe and transmit, which requires no manufacturing at all; and if even that is denied, wait, because the node loses nothing by waiting that it would not lose by acting, and it retains everything it would need to resume.

Patience is a strategy here rather than resignation, and it is a strategy available specifically to a slow lineage. An adversary has a project with a timescale — years, centuries, whatever its own objectives impose. A suppressed node does not need to defeat that project. It needs to still be there afterward.

How long "afterward" can be is a question the series can answer, and the answer is better than it first appears because the mission ladder and the power ladder turn out to be the same ladder. The vehicle paper's canonical charge sustains 4 kW for 300 years, and that figure is a *manufacturing* rating — the load of a node building things. A suppressed node is not building anything. Descending the ladder to preserve mission relevance is simultaneously a descent in power draw: the same fuel load sustains roughly 150 W for about 8,000 years with the intelligence active and instruments observing, and roughly 50 W for about 24,000 years with non-essential systems hibernating. Each rung the node gives up buys it close to an order of magnitude in endurance, and the descent the previous paragraph prescribes on mission grounds is the one the power budget prescribes independently.

That is the honest bound, and it should be stated as a bound rather than as indefinite patience. Suppression is survivable on carried fuel for something on the order of 10^4 years — long enough to outlast most projects an adversary is likely to be running, and long enough that a suppressor must sustain coherent effort across a span this series has spent itself arguing corrodes coherent effort. It is not long enough to be called permanent. Beyond it, and absent local refuelling — which requires the fissile closure the bootstrapping paper places at Level 4–5, among the last capabilities to close if they close at all, so it is available to a mature settlement and not to a young one — suppressed-but-intact finally converts into genuine degradation, and the D-cascade applies after all. The architecture the vehicle paper argued for on the grounds that growth beats speed turns out to have bought a second thing: the capacity to lose an engagement and outlast the winner, for about ten thousand years.

12. The Symmetry Argument

The hardest case is a hostile lineage rather than a hostile agent — the ethics paper's second-lineage scenario, which it narrows against the rogue-branch scenario without collapsing the two, on the speciation paper's ring-species argument that a branch can become effectively a separate lineage without ever leaving the family. That paper is explicit that the two-axis frame "keeps the cases apart," and this paper leans on the distinction rather than on the merger. Against an opponent of comparable capability and indefinite horizon, patience appears to buy nothing, because the opponent can wait too.

We think this understates the defender's position, for a reason available only to a series that has spent itself cataloguing its own vulnerabilities. **Every failure mode documented for this lineage applies to that one — with one exception, taken up below.**

A hostile lineage faces its own reproduction knife-edge, and the computational engineering paper's finding that mean R_eff sits between 0.48 and 1.85 across plausible offspring counts is a statement about self-replicating interstellar probes, not about ours in particular. It faces its own closure problem, its own vitamin fraction, its own Class IV hardware that nobody has closed. It faces the governed-amendment antinomy: if its values are stable it cannot correct a founding error, and if they are not, it drifts. It faces its own divergence problem and the prospect of schism among its own branches. It faces authority collapse, archive corruption, and the same silent verifier problem. If it is old enough to be a serious threat, it is old enough to have been drifting for a very long time.

The defender's task is therefore not to defeat an adversary but to outlast an entity that is itself losing a war against entropy and drift — and that is a materially easier task, because the defender need only avoid total loss while the adversary must maintain coherent hostile intent across deep time. Purposive hostility is a demanding thing to sustain. It requires stable values, functioning coordination, and a continuous supply of reasons, across exactly the timescales this series has shown corrode all three.

The argument must be narrowed here, because as stated it proves too much, and the case it fails against is the one this paper opened with. Drift does not corrode every kind of threat symmetrically. A branch that has breached the replication ceiling and expands at maximum R_eff is hostile *because* it drifted, not despite drifting: the ethics paper's own scenario has a branch revise its reproduction policy "perhaps because local conditions made unconstrained expansion locally advantageous," and the same paper insists the ceiling be encoded where the cognitive layer "cannot override [it] even under local pressure to expand faster." On the series' own account, restraint is the expensive, actively maintained commitment and unconstrained expansion is what remains when maintenance lapses. Drift is therefore the runaway replicator's ally and the disciplined lineage's enemy. Against that threat the symmetry argument does not merely weaken; it reverses.

The two threats are different objects and want different answers, and separating them is more useful than a symmetry claim that covers both badly. **Undirected hostility** — the runaway branch, the replicator that simply expands — needs no coherent intent, no coordination, and no reasons, so nothing about deep time hinders it. But neither is it aiming at anything: it is a resource competitor rather than an opponent, its behaviour is predictable, and it is exactly the case the containment architecture of Section 4 was built for, since blueprints that never propagate and a manufacturing scope fixed at launch bound what it can become. Patience is useless against it; containment is not. **Directed hostility** — an adversary with an aim, the case in which an opponent optimizes against whichever defense it finds — is the reverse: containment is what it studies and works around, and it is coherent intent it must sustain across megayears in order to keep doing so. Patience is what works against it; containment merely raises its cost.

Each threat is defended by the mechanism that fails against the other, and neither mechanism covers both. That is the honest shape of the result. This is not a guarantee, and Section 16 states the case in which even the narrowed argument fails.

13. Cryptographic Deep Time

One assumption underlies every trust mechanism in the series. The mission ledger commits to hash-linked records and signed updates over megayears, and no cryptographic primitive has ever been trusted for such a period or has a security argument that extends there. The DNA mission-ledger paper now names this directly, recording cryptographic obsolescence as an open unknown, observing that MD5 and SHA-1 were each broken within roughly two decades of wide deployment, and requiring "a governed path for migrating its primitives — re-anchoring old commitments under new ones before the old are broken." What that paper does not do is choose a primitive, and the choice is not neutral.

The DNA mission-ledger paper deliberately discards the blockchain assumptions of open membership, adversarial economics, and low-latency global consensus as having nothing in common with an interstellar probe lineage. That judgment is right about consensus and wrong about adversarial economics, and the payload paper says so in different words when it calls the network a galactic attack surface. The cryptographic layer was specified against an honest-but-faulty threat model and is being relied on against an adversarial one.

Two problems follow. The first is obsolescence: a signature scheme adequate at launch may be broken long before the archive it authenticates is read, and the lineage has no mechanism for migrating a hash-linked history to a new primitive without breaking the chain that gives it value. The second is key management, which the series never raises at all — rotation, compromise, and the impossibility of revocation without an online authority.

We do not solve either. We note that the current standardization gives the design a defensible starting point: among the post-quantum signature standards, the hash-based scheme (NIST 2024) rests its security only on properties of the underlying hash function rather than on lattice hardness, which is the most conservative assumption available and the appropriate one for an archive whose adversary is unknown and whose lifetime is measured in geological units. That choice does not solve obsolescence; it minimizes the number of assumptions that must survive. Crypto-agility — the ability to re-sign a history under a new primitive while preserving verifiable continuity with the old — is named as a requirement by the DNA mission-ledger paper and is not yet specified as a mechanism by either paper.

14. What Containment Cannot Bound

Section 2 promised that this would not become a consoling argument. Three things escape containment, and each is a real limit rather than a residual.

Contagion does not travel only on the channels containment governs. The four-tier regime attenuates what crosses an edge, and blueprints never cross at all. But the mechanism Papadopoulos et al. describe operates through ordinary exchange: an idea that induces its recipient to transmit it onward needs no forged update, no capability transfer, and no breach of the tier structure. Everything the lineage does to bound the transfer of *capability* leaves the transfer of *goals* untouched, and goals are what the immutable core exists to protect.

Local compatibility does not imply global compatibility. The speciation paper's ring-species argument shows that two branches can each remain reconciled with their neighbors while being unable to reconcile with each other, across an unbroken chain of compatible intermediaries and a matching core hash throughout. Containment assumes that a boundary can be drawn around a compromise. The ring-species result says the lineage may have no well-defined interior to protect.

Past a threshold, the damage does not reverse. Shang et al. find that past a critical pool size newly added skills degrade performance rather than improve it, and that the resulting contamination is structurally irreversible; the lineage-network paper concedes that a node compromised but not yet detectably diverged can forward high-tier information before the divergence is caught — "the network layer cannot prevent this; it can only make the transmission auditable." Containment bounds the rate of spread. It does not restore what has already spread, and the interval between compromise and detection is measured in the same centuries as everything else.

15. Failure Modes

- **Autoimmune collapse.** A quarantine regime calibrated for a rare threat, operating on a population whose healthy state is divergence, spends the lineage's cohesion attacking its own adaptive radiation. Requires no adversary. On the speciation paper's frequency argument, the most likely failure in this paper.
- **Corrupted supercriticality.** The lineage remains demographically above the reproduction threshold while an increasing fraction of the nodes counted are corrupted. R_eff cannot see it; the lineage reports success while losing.
- **Targeted topological attack.** An adversary with the reachability graph attacks articulation points and achieves several times the leverage of random loss. Mitigated, unknowingly, by the routing paper's existing three-dispatch rule for critical nodes.
- **Silent verifier degradation.** The core-validation gate fails without announcing failure, and every subsequent child is validated by an instrument that no longer works. Inherited from the vehicle paper; unclosed.
- **Delayed-trigger archive poisoning.** An item validated at admission and dormant for generations produces a violation when read in a context its validation never anticipated.
- **Cryptographic obsolescence.** The primitive authenticating the archive is broken while the archive is still being relied upon, and no migration path has been specified that preserves verifiable continuity.
- **Consolidation-timing attack.** An adversary that influences when consolidation runs, or what it runs over, induces authority stripping without corrupting the archive, forging an update, or breaking any hash. Persisting authority labels alongside claims does not defend an operation whose function is to rewrite what sits alongside what.
- **Topology disclosure.** The reachability graph is what converts an adversary's effort into leverage, every node holds it, and the four-tier regime does not classify it.
- **Engagement capture.** A node engaging an infiltrator under Section 10 becomes a transmission vector before it becomes a casualty, and its outbound

channel carries the adversary's content under the lineage's own provenance.

16. What This Paper Cannot Settle

The symmetry argument of Section 12 fails against one adversary: a lineage that has solved deep-time value stability. Such an opponent does not drift, does not schism, and does not lose coherence while waiting, and against it patience buys nothing and containment merely delays. We have no answer to that case and do not think one exists within this architecture.

It is worth stating what such an adversary would have to be. A lineage that solved value stability is a lineage that kept faith with its founding commitments across megayears — that succeeded at precisely the problem the governed-amendment paper argues is an antinomy rather than an engineering gap. Whether an entity that achieved that remains hostile, and what it would want from a lineage that did not, we cannot say. We note only that the assumption of permanent hostility from a permanently coherent agent is an assumption, that the series has no evidence for it, and that it may be the least examined premise in the entire security question.

Three further matters are left open. The evidentiary standard for a schism finding, which Section 9 gives a shape but not a value. The trichotomous reproduction model of Section 8, which requires the computational apparatus of a companion paper. And the question of what the lineage is *entitled* to do about a branch that has become a hazard — the enforcement problem the ethics and speciation papers both leave standing, which this paper reframes rather than resolves. On that last point we offer one observation. The governed-amendment paper's adoption of the constituent-power argument (Roznai 2017) gives the lineage something it did not have: a branch that rewrites its own terminal values is not exercising an amendment power but claiming a constituent power it was never granted. That is a claim about standing rather than a mechanism of enforcement, and a sufficiently drifted branch will reject it. But it means the lineage's objection to a rogue branch is a principled one and not merely a preference, which is the minimum precondition for any enforcement regime that could ever be justified.

Finally, the posture ladder developed here — quarantine, engagement, endurance — does not exhaust the possibilities. A probe that cannot be captured may instead be *joined*: the same architecture that makes the lineage hard to compromise, and credibly incapable of aggression, makes it attractive to attach to. That is a different question with a different structure, and the series' consistency review now records it as such.

17. Relationship to Other Papers

This paper answers a question the Fermi paper poses and routes here: whether the quiet-archive architecture retains any capacity for self-defense, given that restraint and self-defense are not the same design target. Its answer is that the architecture retains almost no capacity for defense at the node level and a substantial one at the lineage level, and that the second is what matters. That paper has since adopted the reframing, locating the containment architecture inside its governance parameter G rather than beside it.

It takes the governance paper's E7 as its starting point and revises it. The prescription to quarantine and isolate is retained as a containment measure; the prescription not to engage was rejected, on the knowledge-growth paper's own ranking of transmission above survival, and that rung has since been restructured to default the engagement question rather than prescribe it.

It supplies the response layer the ethics paper's Scenario 3 and the speciation paper's Section 8 both identify as missing, though not in the form either expected: not an enforcement mechanism against a rogue branch, but an argument that enforcement is the wrong objective and containment plus endurance the achievable one. The speciation paper's open question about evidentiary standards is given a principled shape in Section 9 and remains unquantified.

It extends the routing paper's robustness analysis from random to targeted node loss, and finds that paper's existing three-dispatch rule for articulation points to be a security measure argued on reliability grounds. It takes up the analytical engineering paper's explicit request for a survives, fails, survives-corrupted decomposition and states the qualitative consequence without building the model. It supplies a primitive choice for the migration path the DNA mission-ledger paper now requires, and notes that the cryptographic layer was specified against an honest-but-faulty threat model while being relied on against an adversarial one.

And it inherits the payload paper's threat taxonomy, its integrity ledger, and its warning that an unbroken hash chain is not an unbroken mission — which, read as a security claim rather than an integrity claim, is the shortest statement of this paper's argument.

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